2.1 What Is Diffusion?

Biology · Movement of Substances · Syllabus refs (G2 & G3): 2a

Name: _____ Class: _____ Date: _____

Learning outcomes

- Define diffusion and describe its role in nutrient uptake and gaseous exchange in plants and humans.
- Describe the factors that affect the rate of diffusion.

Trigger activity

Someone sprays perfume at one end of a room. A few moments later, a person at the other end can smell it. **How did the perfume get there, when nobody carried it?**

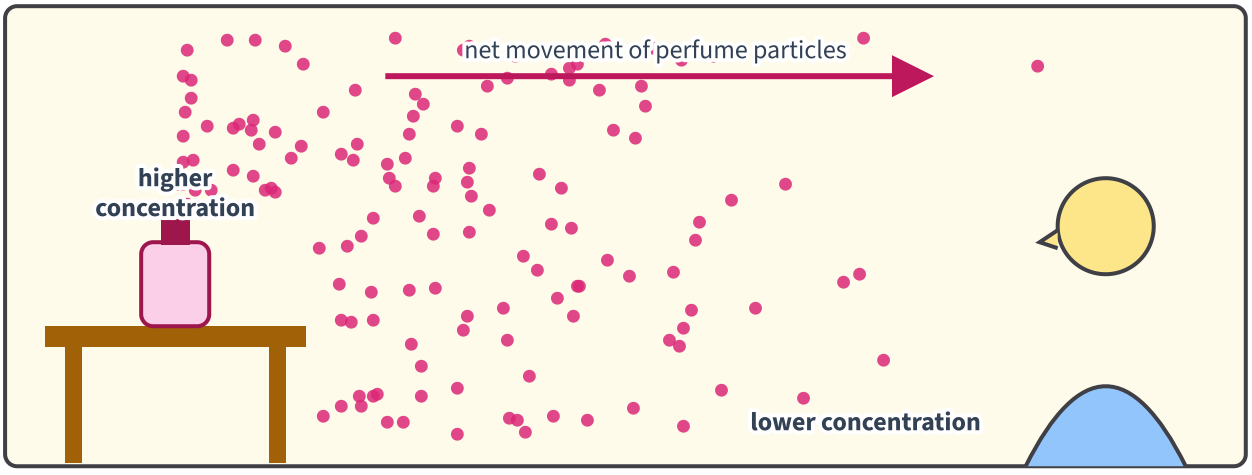


Fig. 2.1 — Perfume particles spread out from where they are crowded to where there are fewer of them.

The perfume particles are always moving about at random, bumping into air particles. Near the bottle there are many perfume particles; far away there are few. So, overall, more particles move *away* from the bottle than towards it. The perfume spreads until it is evenly mixed in the air.

What is diffusion?

The difference in concentration between two regions is called a **concentration gradient**. Particles spread from the region of higher concentration to the region of lower concentration: we say they move **down the concentration gradient**.

Diffusion is the net movement of particles (atoms, molecules or ions) from a region where they are at a higher concentration to a region where they are at a lower concentration, that is, down a concentration gradient.

> **Key idea:** the **steeper** the concentration gradient (the bigger the difference in concentration), the **faster** the rate of diffusion.

Why "net"? Each particle moves at random, in any direction. Some particles even move back towards the crowded region. The *net* movement is the overall result: more particles go one way than the other.

Practical: diffusion of a dissolved substance in water

Aim: to investigate how a dissolved substance spreads through water.

Procedure

1. Gently drop one crystal of **copper(II) sulfate** into a gas jar filled with water. Cover the jar.
2. Leave the jar to stand, without shaking it, for a few days.
3. Observe how the colour of the water changes.

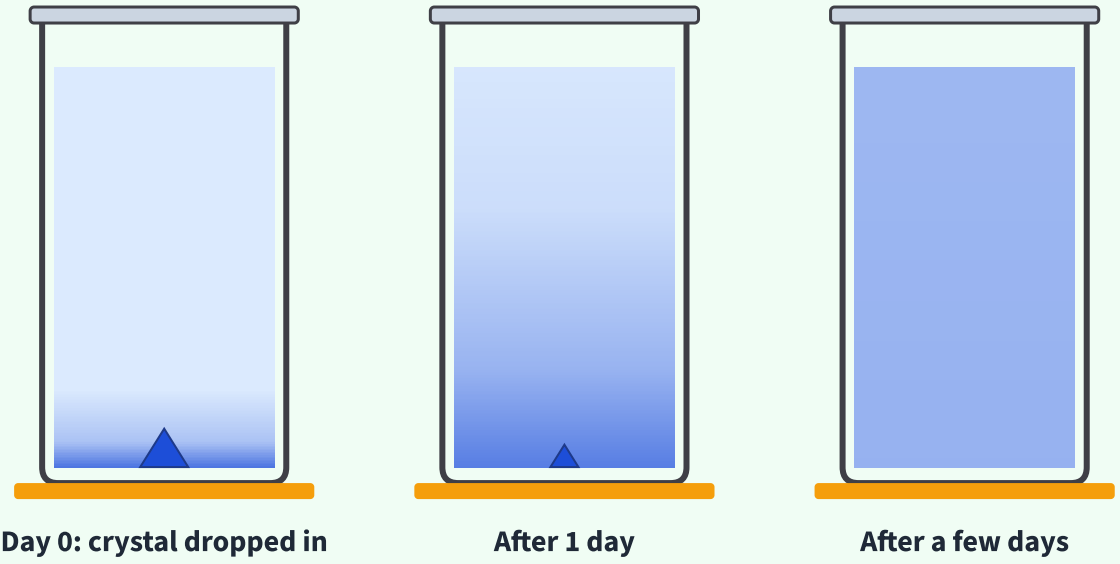


Fig. 2.2 — The copper(II) sulfate crystal dissolves, and the blue colour slowly spreads through all the water.

Observation and discussion

1. Use the idea of a concentration gradient to explain why the blue colour slowly spreads through the water.
2. When does diffusion stop?

In the practical, the copper(II) sulfate particles moved from the region of higher concentration around the crystal to regions of lower concentration in the water. The solute diffused until it was evenly spread, forming a solution.

If **more than one substance** is dissolved in the same liquid, each one diffuses on its own, down its own concentration gradient. The particles of one substance do not affect the diffusion of the other.

Diffusion across a membrane

A **permeable membrane** lets both the solvent (water) and the solutes (the dissolved substances) pass through it. In Fig. 2.3, a permeable membrane divides a beaker into two halves:

- the **left** side has a higher concentration of **potassium nitrate** (yellow);
- the **right** side has a higher concentration of **copper(II) sulfate** (blue).

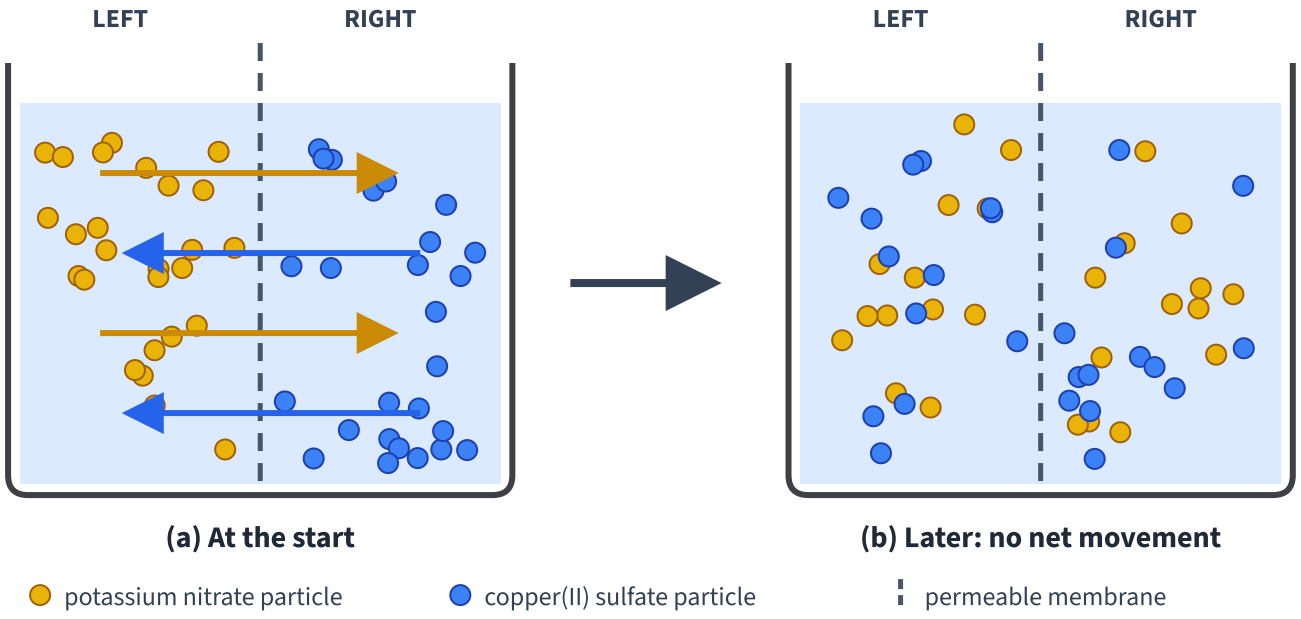


Fig. 2.3 — Potassium nitrate and copper(II) sulfate particles diffusing through a permeable membrane.

So the potassium nitrate particles diffuse across the membrane to the right, and the copper(II) sulfate particles diffuse across to the left. In the end, the concentration of each substance is the **same on both sides** of the membrane.

Factors that affect the rate of diffusion

How fast a substance diffuses depends on:

concentration gradient | diffusion distance | surface area to volume ratio

1. Concentration gradient

The steeper the concentration gradient, the faster the rate of diffusion (see above). When there is no concentration gradient, there is no net diffusion.

2. Diffusion distance

The distance a substance has to travel is the **diffusion distance**. The **shorter** the diffusion distance, the less time the substance needs to get there, so the **faster** the rate of diffusion.

A cell membrane is not like the membrane in Fig. 2.3: it is **partially permeable**. It lets some substances through but not others. Small molecules such as **oxygen** and **carbon dioxide** pass

through easily; large molecules such as **proteins** cannot. Diffusion is an important way for oxygen and carbon dioxide to move into and out of cells.

Diffusion in humans and plants

Gas exchange in the lungs. Oxygen and carbon dioxide are exchanged in the lungs by diffusion. The air in the lungs has a high concentration of oxygen, and the blood arriving there has a low concentration. Oxygen diffuses from the air space, through a very thin wall, into the red blood cells. Carbon dioxide diffuses the other way, from the blood into the air space, and is breathed out.

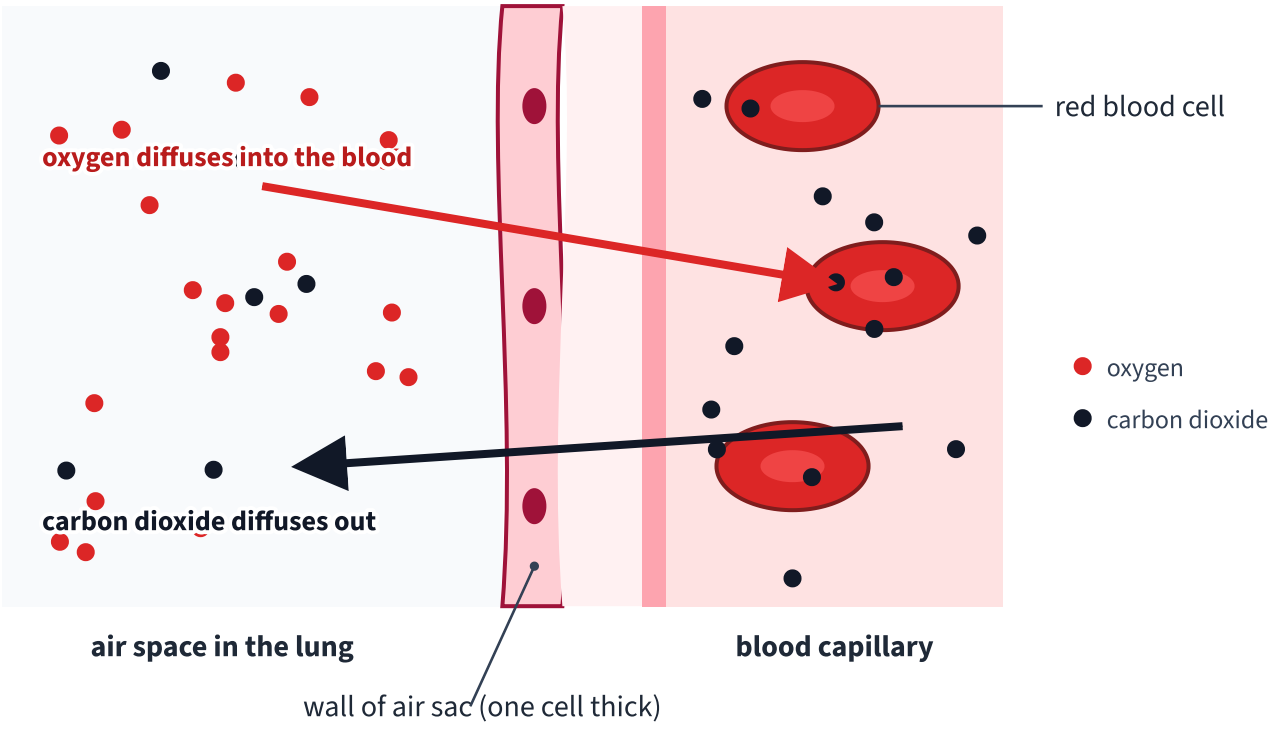


Fig. 2.4(a) — Gas exchange between an air space in the lung and the blood.

Gas exchange in plant cells. Plant cells such as root hair cells also take in oxygen and get rid of carbon dioxide by diffusion. Oxygen diffuses in from the air spaces in the soil, and carbon dioxide diffuses out.

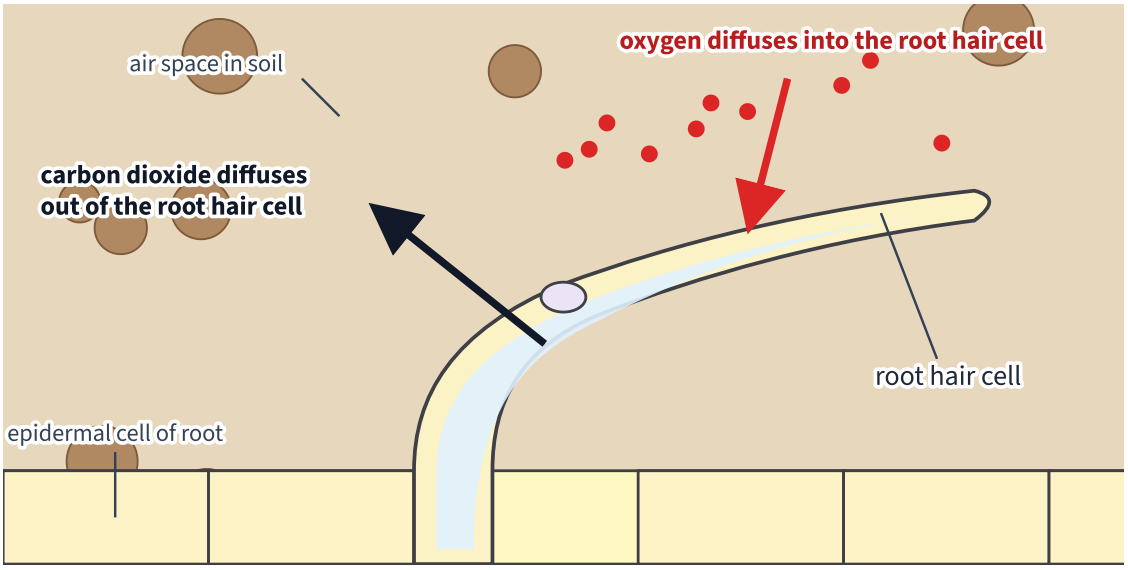


Fig. 2.4(b) — Gas exchange between a root hair cell and the air in the soil.

In both cases the diffusion distance is **short**, so diffusion happens quickly.

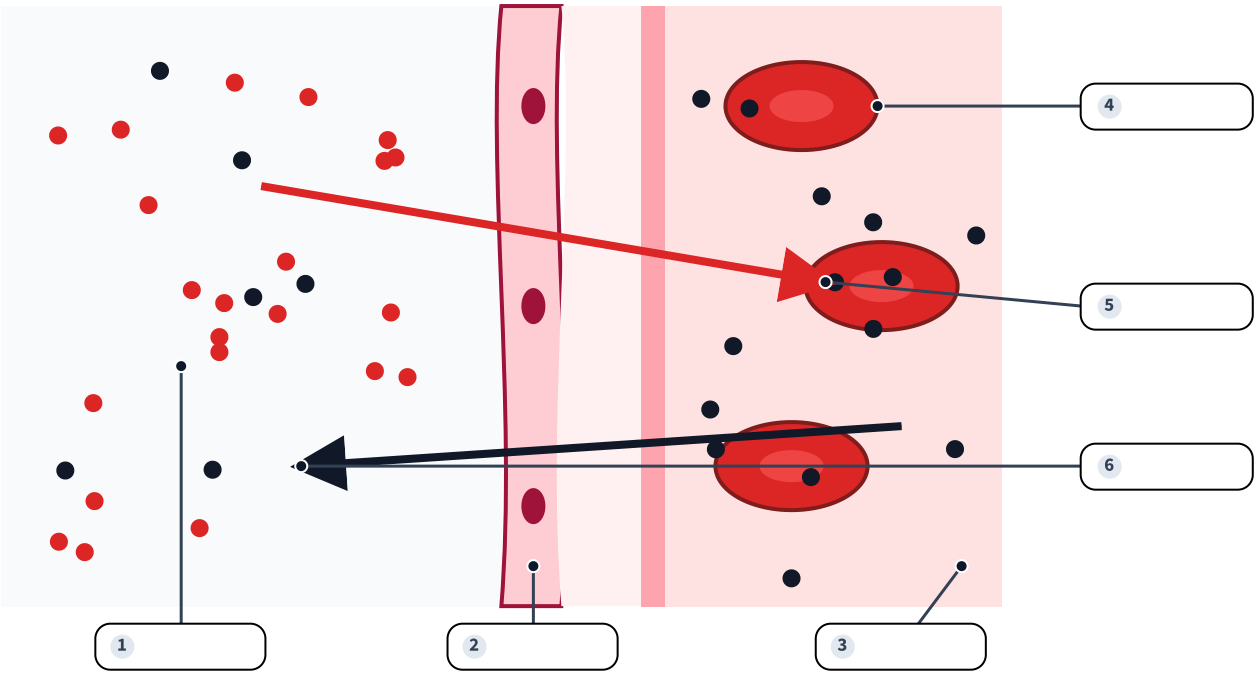
Taking in nutrients. Dissolved substances (solutes) can diffuse too. In the **small intestine**, digested food such as **glucose** and **amino acids** diffuses into the blood capillaries in the wall of the intestine, and the blood carries it around the body. Root hair cells take in some mineral salts from the soil in the same way.

> A **solute** is a substance that is dissolved in another substance (the **solvent**). Together they make a **solution**.

> **Big idea: systems**
>
> Living things depend on diffusion to exchange gases with their surroundings. Without it, cells would not get the oxygen they need to stay alive. You will meet gas exchange again in the topics on respiration in humans and on plants.

Label gas exchange in the lungs.

chloroplast | wall of air sac | oxygen | red blood cell | blood capillary | air space in the lung | cell wall | carbon dioxide



Gas exchange in the lungs.

3. Surface area to volume ratio

Why are cells usually so small? Why are large organisms made of many small cells instead of one giant cell?

The answer is about how fast materials can cross the cell membrane. A cell must take in oxygen and nutrients, and get rid of waste, fast enough to stay alive.

Substances enter and leave a cell through its surface, so the rate depends on how big the cell membrane is: the cell's **surface area**. For two cells with the same volume, the one with the larger surface area takes substances in faster. Comparing a cell's surface area with its volume gives its **surface area to volume ratio** (SA : V).

> **Key idea:** the **greater** the surface area to volume ratio of a cell, the **faster** substances move into and out of it.

As a cell gets bigger, its volume grows faster than its surface area, so its SA : V gets smaller. A very large cell could not take in oxygen fast enough for all of its volume. That is why cells stay small,

and why specialised cells such as root hair cells and red blood cells have shapes that give them a large SA : V.

Try it again: the shape builder and the agar-cube experiment on SA : V are in Lesson 1.2.

A. Cube X has sides of 1 cm: surface area 6 cm^2, volume 1 cm^3. Cube Y has sides of 3 cm: surface area 54 cm^2, volume 27 cm^3. Work out the SA : V of cube Y (as a number : 1).

Your answer (: 1)

Check your understanding

1. Which statement describes diffusion?

- ○ Net movement of particles from a region of lower concentration to a region of higher concentration
- ○ Net movement of particles from a region of higher concentration to a region of lower concentration
- ○ Movement of water only, through a partially permeable membrane
- ○ Movement of particles only through a membrane

2. A crystal of copper(II) sulfate is left in a gas jar of water. After many days the water is evenly blue. Which is correct?

- ○ The copper(II) sulfate particles have stopped moving.
- ○ There is no net movement, but the particles are still moving.
- ○ The particles are now moving towards the bottom of the jar.
- ○ The water has turned into copper(II) sulfate.

3. Which change makes diffusion faster?

- ○ A shallower concentration gradient
- ○ A longer diffusion distance
- ○ A smaller surface area to volume ratio
- ○ A steeper concentration gradient

4. In the lungs, in which direction does oxygen diffuse?

- ○ From the blood into the air space
- ○ From the air space into the blood
- ○ It does not move
- ○ From the red blood cells into the air space

5. Which substance cannot pass through a cell membrane?

- ○ Oxygen
- ○ Carbon dioxide
- ○ Water
- ○ Protein

6. How do glucose and amino acids get from the small intestine into the blood?

- ○ By diffusion into the blood capillaries
- ○ By photosynthesis
- ○ By breathing
- ○ They stay in the small intestine

7. Why are most cells very small?

- ○ Small cells have a smaller surface area to volume ratio.
- ○ Small cells have a larger surface area to volume ratio, so substances can move in and out fast enough.
- ○ Small cells do not need oxygen.
- ○ Small cells have no cell membrane.

8. A beaker is divided by a permeable membrane. The left side has more potassium nitrate; the right side has more copper(II) sulfate. What happens?

- ○ Only potassium nitrate moves.
- ○ Potassium nitrate moves to the right and copper(II) sulfate moves to the left.
- ○ Both substances move to the left.
- ○ Nothing moves, because they block each other.